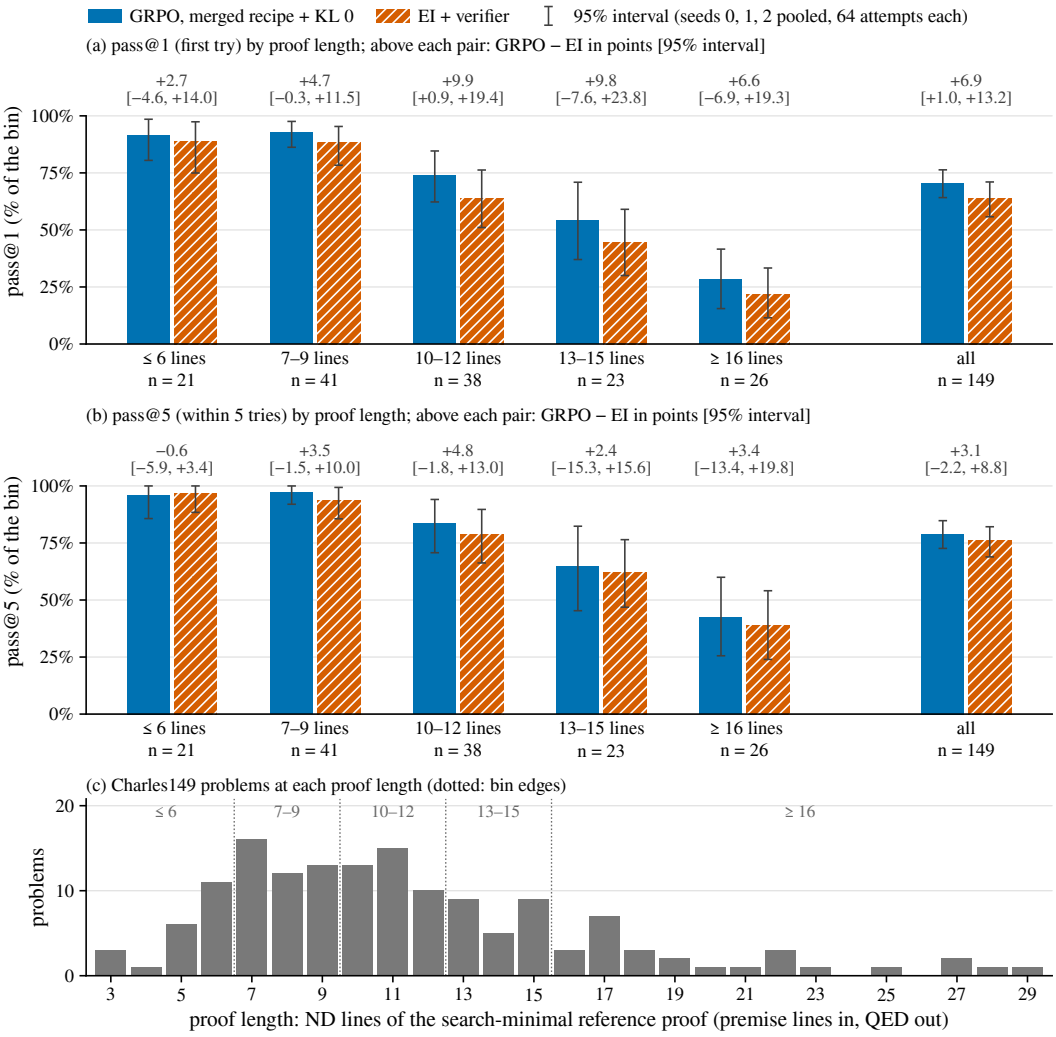
GRPO, merged recipe + KL 0
EI + verifier
95% interval (seeds 0, 1, 2 pooled, 64 attempts each)
(a) pass@1 (first try) by proof length; above each pair: GRPO − EI in points [95% interval]
+2.7 [−4.6, +14.0]
+4.7 [−0.3, +11.5]
+9.9 [+0.9, +19.4]
+9.8 [−7.6, +23.8]
+6.6 [−6.9, +19.3]
+6.9 [+1.0, +13.2]
pass@1 (% of the bin)
100%
75%
50%
25%
0%
≤ 6 lines n = 21
7–9 lines n = 41
10–12 lines n = 38
13–15 lines n = 23
≥ 16 lines n = 26
all n = 149
(b) pass@5 (within 5 tries) by proof length; above each pair: GRPO − EI in points [95% interval]
−0.6 [−5.9, +3.4]
+3.5 [−1.5, +10.0]
+4.8 [−1.8, +13.0]
+2.4 [−15.3, +15.6]
+3.4 [−13.4, +19.8]
+3.1 [−2.2, +8.8]
pass@5 (% of the bin)
100%
75%
50%
25%
0%
≤ 6 lines n = 21
7–9 lines n = 41
10–12 lines n = 38
13–15 lines n = 23
≥ 16 lines n = 26
all n = 149
(c) Charles149 problems at each proof length (dotted: bin edges)
problems
20
10
0
≤ 6
7–9
10–12
13–15
≥ 16
3
5
7
9
11
13
15
17
19
21
23
25
27
29
proof length: ND lines of the search-minimal reference proof (premise lines in, QED out)